

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the Credit Card Approval Prediction System.

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Description	Priority
1	Authentication	Secure login for bank analysts and compliance officers.	High
2	Authorization	Role-based access for Analysts (dashboard) vs Customers (sandbox).	High
3	External Interfaces	IBM Watson ML integration and Flask web interface.	High
4	Transaction Processing	Real-time prediction of approval/rejection outcomes.	High
5	Feature Engineering	Binary risk labeling (High/Low) and data scaling.	High
6	Explainability	Generate SHAP values and feature importance plots per case.	Medium
7	Notification System	Alert analysts for borderline application cases.	Medium
8	Customer Sandbox	Simulation mode for customers to test financial scenarios.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	Fast real-time model prediction.	< 5 seconds

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2	Scalability	Handle request loads on IBM Cloud.	10,000+ per hour
3	Security & Privacy	Encrypt applicant financial profiles.	AES-256
4	Reliability & Availability	Cloud-based uptime for analysts.	99.5% uptime
5	Usability	Intuitive Dashboard with Matplotlib charts.	User friendly
6	Portability	Deployment on IBM Watson and Flask local.	Cross-platform